

Lab 4 - Multinomial Regression

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Lab Goal: Predict voting frequency using demographic variables

Data source: FiveThirtyEight “Why Many Americans Don’t Vote” survey

Method: Multinomial logistic regression

0.1 Instructions

- If you are fitting a model, display the model output in a neatly formatted table. (The `tidy()` and `kable()` functions can help!)
- If you are creating a plot, use clear labels for all axes, titles, etc.
- Commit and push your work to GitHub regularly, at least after each exercise. Write short and informative commit messages.
- When you’re done, we should be able to knit the final version of the QMD in your GitHub as an HTML.

Workflow tip

Work through the lab by running chunks **one at a time** in RStudio (Cmd/Ctrl + Enter), not by rendering the whole document. Render only at the end, once all your code is working.

This is how most R users actually work – you’d write a line, run it, check the output, and iterate. Rendering is for producing the final document. When you render, Quarto starts a fresh R session and runs every chunk top-to-bottom, so if anything is incomplete, everything downstream breaks. When you work interactively using chunks, your environment builds up as you go, so you always have your objects available to inspect and troubleshoot.

The “**Run All Chunks Above**” button (downward arrow with a bar, next to the green play button in each chunk) is another useful tool. It re-runs everything from the top of the document up to your current chunk, so all your objects are up to date before you start working on the next question.

0.2 Setup

```
library(nnet)
library(car)
library(tidyverse)
library(emmeans)
library(ggeffects)
library(knitr)
library(patchwork)
library(broom)
library(parameters)
library(easystats)
library(viridis)
```

1 Part 1: Data Preparation

1.1 The Data

The data for this assignment comes from an online Ipsos survey that was conducted for the FiveThirtyEight article [“Why Many Americans Don’t Vote”](#). If you’d like, you can read more about the survey design and respondents in the README of the [GitHub repo](#) for the data, but all the necessary information is explained below.

Respondents were asked a variety of questions about their political beliefs, thoughts on multiple issues, and voting behavior. We will focus on using the demographic variables and someone’s party identification to understand whether a person is a probable voter.

The variables we’ll focus on are (definitions from the codebook in data set GitHub repo):

- **ppage**: Age of respondent
- **educ**: Highest educational attainment category
- **race**: Race of respondent, census categories. Note: all categories except Hispanic were non-Hispanic.
- **gender**: Gender of respondent
- **income_cat**: Household income category of respondent
- **Q30**: Response to the question “Generally speaking, do you think of yourself as a...”

- 1: Republican
 - 2: Democrat
 - 3: Independent
 - 4: Another party, please specify
 - 5: No preference
 - -1: No response
- **voter_category**: past voting behavior:
 - **always**: respondent voted in all or all-but-one of the elections they were eligible in
 - **sporadic**: respondent voted in at least two, but fewer than all-but-one of the elections they were eligible in
 - **rarely/never**: respondent voted in 0 or 1 of the elections they were eligible in

```
voter_data <- read_csv("https://raw.githubusercontent.com/fivethirtyeight/data/master/non-voter")
```

The variable Q30 contains the respondent's political party identification. The following code simplifies Q30 into four categories:

```
voter_data <- voter_data %>%
  mutate(pol_ident = case_when(
    Q30 == 1 ~ "Rep",
    Q30 == 2 ~ "Dem",
    Q30 == 3 ~ "Indep",
    TRUE ~ "Other"
  ))
```

Question 1.1

The variable **voter_category** identifies the respondent's past voter behavior. Relevel the variable to make **rarely/never** the baseline level, followed by sporadic, then always.

Hint

Use **factor()** with the **levels** argument to set the ordering. The first level becomes the baseline in **multinom()**.

```
# Enter your code
```

Question 1.2

Mean-center the age variable (`ppage`) so that the intercept reflects the log-odds for an average-aged person rather than a 0-year-old.

```
# Enter your code
```

2 Part 2: Exploratory Data Analysis

Question 2.1

In the [FiveThirtyEight article](#), the authors include visualizations of the relationship between voter category and demographic variables such as race, age, education, etc.

Select **two** demographic variables. For each variable, create a visualization showing its relationship with `voter_category`. Combine them into a single figure using `patchwork`. Briefly describe (1–2 sentences each) what the plots reveal.

For inspiration on recreating the FiveThirtyEight style: <https://www.mikelee.co/posts/2020-02-08-recreate-fivethirtyeight-chicklet-stacked-bar-chart-in-ggplot2>

💡 Hint: Quick proportional bar charts

`geom_bar(position = "fill")` gives proportions instead of counts. Map `x` to your demographic variable and `fill` to `voter_category`.

```
# Plot 1
# plot1 <- ggplot(...)
```

```
# Plot 2
# plot2 <- ggplot(...)
```

```
# Combine with patchwork
# plot1 + plot2
```

Enter your interpretation here: _____

3 Part 3: Model Building

Question 3.1

Fit a multinomial logistic regression model using mean-centered age, race, gender, income, and education to predict `voter_category`. Save this as `model_demo`.

You don't need to display the output here — we'll examine it in Part 4. If you want to peek at it, feel free to run `summary(model_demo)` in your console. There's no real harm in displaying it via the code chunk, except that it'll make your rendered page quite messy. `kable()` is better for clean tables.

Hint

```
multinom(voter_category ~ age_centered + race + gender + income_cat +  
educ, data = voter_data, trace = FALSE) — the trace = FALSE suppresses the  
iteration output.
```

To keep the rendered document clean, add `#| results: hide` to the chunk header — this runs the code but hides the output. You can always check the output interactively by running the chunk in the console.

```
# Enter your code
```

Question 3.2

Should party identification (`pol_ident`) be added to the model? Fit a second model that adds `pol_ident` to the predictors. Use `anova()` to compare the two models. Report the test statistic and your conclusion.

```
# Enter your code
```

Enter your answer: _____

Use the model you select for the remainder of the assignment. For clarity, call your chosen model `final_model`.

4 Part 4: Interpreting the Model

Question 4.1: Overall significance

Use a likelihood ratio test (`car::Anova()` with `type = "III"`) to report the overall significance of each predictor in `final_model`.

```
# Enter your code
```

Question 4.2: Raw coefficients

Display the model output using `tidy(final_model) %>% kable(digits = 3)`.

Pick **one** comparison (e.g., “sporadic vs. rarely/never”) and do the following:

- Write out the model equation for that comparison.
- Interpret the coefficient for mean-centered age: what happens to the **odds** of being in that category (vs. the baseline) for each additional year of age? Report the odds ratio `exp()`.

 Hint: Reading the `tidy()` output

When you run `tidy(final_model)`, you’ll see a table with columns like `y.level`, `term`, `estimate`, etc. The table has **two blocks of rows** — one block where `y.level` is `sporadic`, and another where `y.level` is `always`. Each block is a separate logistic regression equation:

- The `sporadic` rows are the equation for: log-odds of `sporadic` **vs. rarely/never**
- The `always` rows are the equation for: log-odds of `always` **vs. rarely/never**

Pick one of those two blocks. The `(Intercept)` row is β_0 and the `age_centered` row is β_1 for that equation.

```
# Enter your code
```

Enter your equation and interpretation here: _____

Question 4.3: Changing the baseline

Refit the model with a **different** baseline category for `voter_category` (e.g., “always” instead of “rarely/never”). Then predict the probabilities for a specific observation using both models.

Do the coefficients change? Do the predicted probabilities change? Explain in 1–2 sentences why this happens.

Hint

Use `relevel(voter_category, ref = "always")` before refitting. Use `predict(model, newdata = ..., type = "probs")` to get predicted probabilities. You can use `head(voter_data, 1)` as your newdata for a quick check.

Enter your code

Enter your answer: _____

5 Part 5: Marginal Effects & Visualization

Question 5.1: Predicted probabilities by age

Plot the predicted probabilities of voter category as a function of age. The code below will work once you replace `final_model` with the name of your model:

Note

This chunk is set to `eval: false` so the file renders before you’ve built your model. Once `final_model` exists in your environment, delete the `#| eval: false` line so the chunk runs when you render.

```
gggemmeans(final_model, terms = c("ppage [all]")) %>%
  ggplot(., aes(x = x, y = predicted, fill = response.level)) +
  geom_area() +
  geom_rug(sides = "b", position = "jitter", alpha = .2) +
  labs(x = "\nAge", y = "Predicted Probability\n",
       title = "Predicted Probabilities of Voting Frequency by Age") +
  scale_fill_manual(
    name = NULL,
```

```

values = c("always" = "#F6B533", "sporadic" = "#D07EA2", "rarely/never" = "#9854F7"),
labels = c("RARELY OR NEVER VOTE", "SOMETIMES VOTE", "ALMOST ALWAYS VOTE"),
breaks = c("rarely/never", "sporadic", "always")
) +
theme_minimal()

```

Describe the pattern you see in 1–2 sentences.

Enter your interpretation here: _____

Question 5.2: Predicted probabilities by education

Create a similar plot showing predicted probabilities of voter category as a function of **education level**.

Hint

Replace "ppage [all]" with "educ" in the `ggemmeans()` call. Since education is categorical, you may want to use `geom_col(position = "dodge")` or `geom_col(position = "fill")` instead of `geom_area()`.

Enter your code

Enter your interpretation here: _____

6 Part 6: Pairwise Comparisons

Question 6.1: Political group differences

Run the following code (replacing `final_model` with your model name) and interpret the output. The code computes estimated marginal means for each political group within each voter category, then tests pairwise differences on the log-odds scale.

Note

Same as Q5.1 — this chunk is set to `eval: false`. Delete that line once `final_model` exists.


```
# Estimated marginal means: political group by voter category
multi_an_pol <- emmeans(final_model, ~ pol_ident | voter_category)

# Contrasts on the log scale (log-odds differences between voter categories, within each pol.
coefs_pol <- contrast(regrid(multi_an_pol, "log"), "trt.vs.ctrl1", by = "pol_ident")

# Display results, organized by contrast
update(coefs_pol, by = "contrast") %>%
  kable(format = "markdown", digits = 3)

# Pairwise differences between political groups, within each contrast
contrast(coefs_pol, "revpairwise", by = "contrast") %>%
  kable(format = "markdown", digits = 3)
```

Enter your interpretation here: _____

Question 6.2: Education level differences

Adapt the code from 6.1 to examine differences across **education levels** instead of political groups. Run it and interpret.

 Hint

Replace `pol_ident` with `educ` in the `emmeans()` and `contrast()` calls.

```
# Enter your code
```

Enter your interpretation here: _____

7 Part 7: Write-Up

Question 7.1

Write a **1-paragraph results section** for your final model. Your paragraph should include:

- The model type and baseline category
- Sample size
- At least one key coefficient reported as an odds ratio with a confidence interval

- The result of the likelihood ratio test for at least one predictor
- At least one finding translated into predicted probability terms (from your plots)

 Example write-up (from a different dataset)

We used a multinomial logistic regression model (baseline category: rarely/never) to predict voting frequency from age, race, gender, income, education, and political identification ($N = [n]$). The likelihood ratio test indicated that political identification was a significant predictor of voter category, $\chi^2(df) = [value]$, $p < .001$. Among [comparison], each additional year of age was associated with [an increase/decrease] in the odds of being [category] vs. rarely/never voting ($OR = [value]$, 95% CI [[lower], [upper]]). Predicted probability plots showed that [describe key pattern from your visualizations].

Enter your write-up here: _____